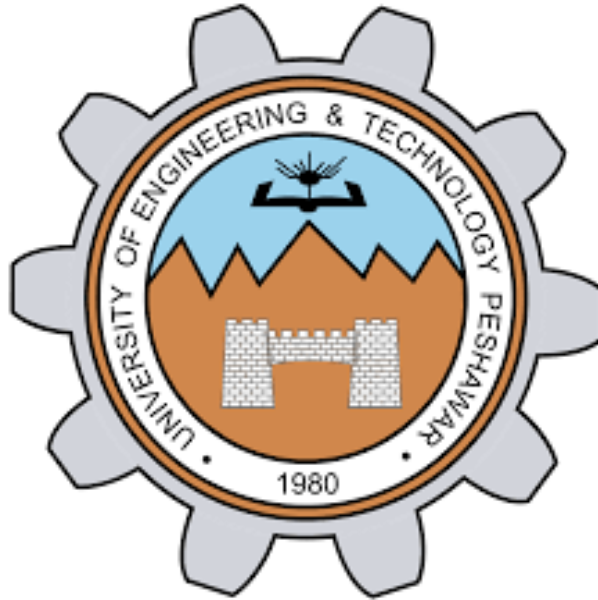




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DEPARTMENT: ELECTRICAL ENGINEERING (POWER)

SECTION: A

SUBJECT: CPP

ASSIGNMENT NO#: 02

SUBMITTED TO: SIR PROFESSOR RIZWAN .

TASK 01 : Find Maximum, Minimum, And Average.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int arr[10];
```

```
    int max, min;
```

```
    float sum = 0;
```

```
    cout << "Enter 10 integers:\n";
```

```
    for(int i=0;i<10;i++) {
```

```
        cin >> arr[i];
```

```
    }
```

```
    max = min = arr[0];
```

```
    for(int i=0;i<10;i++) {
```

```
        if(arr[i] > max) max = arr[i];
```

```
        if(arr[i] < min) min = arr[i];
```

```
        sum += arr[i];
```

```
    }
```

```
    cout << "Maximum = " << max << endl;
```

```
    cout << "Minimum = " << min << endl;
```

```
    cout << "Average = " << sum/10 << endl;
```

```
    return 0;
}
```

TASK 02 : REVERSE A 1D ARRAY USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
void reverseArray(int arr[], int size) {
```

```
    for(int i=0;i<size/2;i++) {
```

```
        int temp = arr[i];
```

```
        arr[i] = arr[size-i-1];
```

```
        arr[size-i-1] = temp;
```

```
    }
```

```
}
```

```
int main() {
```

```
    int arr[5];
```

```
    cout << "Enter 5 elements:\n";
```

```
    for(int i=0;i<5;i++) cin >> arr[i];
```

```
    cout << "Before Reverse:\n";
```

```
    for(int i=0;i<5;i++) cout << arr[i] << " ";
```

```
    reverseArray(arr,5);
```

```
cout << "\nAfter Reverse:\n";

for(int i=0;i<5;i++) cout << arr[i] << " ";

return 0;

}
```

TASK 03: MATRIX ADDITION.

```
#include <iostream>

using namespace std;

int main() {

    int A[3][3], B[3][3], C[3][3];

    cout << "Enter Matrix A:\n";

    for(int i=0;i<3;i++)

        for(int j=0;j<3;j++)

            cin >> A[i][j];

    cout << "Enter Matrix B:\n";

    for(int i=0;i<3;i++)

        for(int j=0;j<3;j++)

            cin >> B[i][j];

    for(int i=0;i<3;i++)

        for(int j=0;j<3;j++)

            C[i][j] = A[i][j] + B[i][j];
```

```
cout << "Result Matrix:\n";

for(int i=0;i<3;i++) {

    for(int j=0;j<3;j++)

        cout << C[i][j] << " ";

    cout << endl;

}


return 0;

}
```

TASK 04: SORT NAMES ALPHABATICALLY.

```
#include <iostream>

using namespace std;

int main() {

    string names[5];


    cout << "Enter 5 names:\n";

    for(int i=0;i<5;i++) cin >> names[i];


    for(int i=0;i<5;i++) {

        for(int j=i+1;j<5;j++) {

            if(names[i] > names[j]) {

                string temp = names[i];

                names[i] = names[j];
```

```

        names[j] = temp;
    }

}

}

cout << "Sorted Names:\n";

for(int i=0;i<5;i++) cout << names[i] << endl;

return 0;

}

```

TASK 05 : POWER SYSTEM LOAD ANALYSIS USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
float totalLoad(float arr[]) {

    float sum = 0;

    for(int i=0;i<24;i++) sum += arr[i];

    return sum;

}

```

```
float peakLoad(float arr[]) {

    float max = arr[0];

    for(int i=1;i<24;i++)

        if(arr[i] > max) max = arr[i];

    return max;
}

```

```
}
```

```
int main() {
```

```
    float load[24];
```

```
    cout << "Enter load for 24 hours:\n";
```

```
    for(int i=0;i<24;i++) cin >> load[i];
```

```
    cout << "Total Load = " << totalLoad(load) << endl;
```

```
    cout << "Peak Load = " << peakLoad(load) << endl;
```

```
    return 0;
```

```
}
```

TASK 06: STUDENTS RECORD SYSTEM USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
int findTopper(int marks[], int size) {
```

```
    int maxIndex = 0;
```

```
    for(int i=1;i<size;i++)
```

```
        if(marks[i] > marks[maxIndex]) maxIndex = i;
```

```
    return maxIndex;
```

```
}
```

```
void passFail(int marks[], int size) {
```

```
for(int i=0;i<size;i++) {  
    if(marks[i] >= 50)  
        cout << "Pass\n";  
    else  
        cout << "Fail\n";  
}  
}
```

```
int searchStudent(string names[], int size, string key) {  
    for(int i=0;i<size;i++)  
        if(names[i] == key) return i;  
    return -1;  
}
```

```
int main() {  
    string names[5];  
    int marks[5];  
  
    cout << "Enter names and marks:\n";  
    for(int i=0;i<5;i++)  
        cin >> names[i] >> marks[i];  
  
    int topper = findTopper(marks,5);  
    cout << "Topper = " << names[topper] << endl;
```



```

passFail(marks,5);

string key;

cout << "Enter name to search: ";

cin >> key;

int index = searchStudent(names,5,key);

if(index != -1)

    cout << "Marks = " << marks[index];

else

    cout << "Student not found";

return 0;

}

```

TASK 07: RENEWABLE ENERGY MONITORING USING FUNTIONS.

```

#include <iostream>

using namespace std;

float totalEnergy(float arr[]) {

    float sum = 0;

    for(int i=0;i<24;i++) sum += arr[i];

    return sum;

}

int maxHour(float arr[]) {

```

```

int index = 0;

for(int i=1;i<24;i++)

    if(arr[i] > arr[index]) index = i;

return index;
}

```

```

float avgEnergy(float arr[]) {

    return totalEnergy(arr)/24;

}

```

```

int main() {

    float energy[24];

    cout << "Enter energy for 24 hours:\n";

    for(int i=0;i<24;i++) cin >> energy[i];

    cout << "Total Energy = " << totalEnergy(energy) << endl;

    cout << "Hour with Maximum Generation = " << maxHour(energy) << endl;

    cout << "Average Energy = " << avgEnergy(energy) << endl;

    return 0;

}

```

TASK 08 : BATTERY HEALTH MONITORING USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
int countBelow(float arr[]) {  
    int count = 0;  
    for(int i=0;i<20;i++)  
        if(arr[i] < 3.0) count++;  
    return count;  
}
```

```
int countAbove(float arr[]) {  
    int count = 0;  
    for(int i=0;i<20;i++)  
        if(arr[i] > 4.2) count++;  
    return count;  
}
```

```
float findMin(float arr[]) {  
    float min = arr[0];  
    for(int i=1;i<20;i++)  
        if(arr[i] < min) min = arr[i];  
    return min;  
}
```

```
float findMax(float arr[]) {  
    float max = arr[0];
```

```

    for(int i=1;i<20;i++)
        if(arr[i] > max) max = arr[i];
    return max;
}

int main() {
    float voltage[20];

    cout << "Enter 20 voltage readings:\n";
    for(int i=0;i<20;i++) cin >> voltage[i];

    int below = countBelow(voltage);
    int above = countAbove(voltage);

    cout << "Below Safe Limit = " << below << endl;
    cout << "Above Safe Limit = " << above << endl;
    cout << "Minimum Voltage = " << findMin(voltage) << endl;
    cout << "Maximum Voltage = " << findMax(voltage) << endl;

    if(below > 0 || above > 0)
        cout << "Warning: Unsafe readings exist!\n";

    return 0;
}

```

TASK 09: SMART GRID DISTRIBUTION USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
void regionTotal(int arr[4][7]) {
```

```
    for(int i=0;i<4;i++) {
```

```
        int sum=0;
```

```
        for(int j=0;j<7;j++) sum += arr[i][j];
```

```
        cout << "Region " << i+1 << " Total = " << sum << endl;
```

```
    }
```

```
}
```

```
int highestDay(int arr[4][7]) {
```

```
    int maxDay=0, maxSum=0;
```

```
    for(int j=0;j<7;j++) {
```

```
        int sum=0;
```

```
        for(int i=0;i<4;i++) sum += arr[i][j];
```

```
        if(sum > maxSum) {
```

```
            maxSum = sum;
```

```
            maxDay = j;
```

```
        }
```

```
    }
```

```
    return maxDay;
```

```
}
```

```
float overallAverage(int arr[4][7]) {
```

```
    int total=0;
```

```
    for(int i=0;i<4;i++)
```

```
        for(int j=0;j<7;j++)
```

```
            total += arr[i][j];
```

```
    return total / 28.0;
```

```
}
```

```
int main() {
```

```
    int power[4][7];
```

```
    cout << "Enter data:\n";
```

```
    for(int i=0;i<4;i++)
```

```
        for(int j=0;j<7;j++)
```

```
            cin >> power[i][j];
```

```
    regionTotal(power);
```

```
    cout << "Highest Demand Day = " << highestDay(power)+1 << endl;
```

```
    cout << "Overall Average = " << overallAverage(power) << endl;
```

```
    return 0;
```

```
}
```

TASK 10 : CLASSROOM ATTENDANCE SYSTEM USING FUNTIONS.

```
#include <iostream>
```

```
using namespace std;
```

```
void studentAttendance(int arr[5][7]) {
```

```
    for(int i=0;i<5;i++) {
```

```
        int sum=0;
```

```
        for(int j=0;j<7;j++) sum += arr[i][j];
```

```
        cout << "Student " << i+1 << " Attendance = " << sum << endl;
```

```
    }
```

```
}
```

```
int highestAttendance(int arr[5][7]) {
```

```
    int maxStudent=0, maxDays=0;
```

```
    for(int i=0;i<5;i++) {
```

```
        int sum=0;
```

```
        for(int j=0;j<7;j++) sum += arr[i][j];
```

```
        if(sum > maxDays) {
```

```
            maxDays = sum;
```

```
            maxStudent = i;
```

```
        }
```

```
    }
```

```
    return maxStudent;
```

```
}
```

```
float overallPercentage(int arr[5][7]) {
```

```
    int total=0;
```

```
    for(int i=0;i<5;i++)
```

```
        for(int j=0;j<7;j++)
```

```
            total += arr[i][j];
```

```
    return (total / 35.0) * 100;
```

```
}
```

```
int main() {
```

```
    int att[5][7];
```

```
    cout << "Enter attendance (1=Present, 0=Absent):\n";
```

```
    for(int i=0;i<5;i++)
```

```
        for(int j=0;j<7;j++)
```

```
            cin >> att[i][j];
```

```
    studentAttendance(att);
```

```
    cout << "Highest Attendance Student = " << highestAttendance(att)+1 << endl;
```

```
    cout << "Overall Attendance Percentage = " << overallPercentage(att) << "%" << endl;
```

```
    return 0;
```

```
}
```